```
#==============================
#-GROUP PROJECT- TEAM 3- FILE UPLOAD
#============================
from google.colab import files
uploaded = files.upload()
```

Chọn tệp Không có tệp nào được chọn Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving netflix_customer_churn.csv to netflix_customer_churn.csv
```

CLEANING DATA

```
import pandas as pd
# Load the dataset
df = pd.read_csv(
    "netflix_customer_churn.csv",
    engine='python',
    encoding="utf-8",
    on_bad_lines="skip"
    )

df.head()
```

	customer_id	age	gender	subscription_type	watch_hours	last_login_days	region	device	monthly_fee	churned
0	a9b75100-82a8-427a-a208-72f24052884a	51	Other	Basic	14.73	29	Africa	TV	8.99	1
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	47	Other	Standard	0.70	19	Europe	Mobile	13.99	1
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	27	Female	Standard	16.32	10	Asia	TV	13.99	0
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	53	Other	Premium	4.51	12	Oceania	TV	17.99	1
4	4e265c34-103a-4dbb-9553-76c9aa47e946	56	Other	Standard	1.89	13	Africa	Mobile	13.99	1

```
df.shape
```

```
(5000, 14)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'subscription_type', 'watch_hours',
       'last_login_days', 'region', 'device', 'monthly_fee', 'churned',
       'payment_method', 'number_of_profiles', 'avg_watch_time_per_day',
       'favorite_genre'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             5000 non-null   object
 1   age                     5000 non-null   int64
 2   gender                  5000 non-null   object
 3   subscription_type       5000 non-null   object
 4   watch_hours             5000 non-null   float64
 5   last_login_days         5000 non-null   int64
 6   region                  5000 non-null   object
 7   device                  5000 non-null   object
 8   monthly_fee             5000 non-null   float64
 9   churned                 5000 non-null   int64
 10  payment_method          5000 non-null   object
 11  number_of_profiles      5000 non-null   int64
```

```
 12  avg_watch_time_per_day  5000 non-null   float64
 13  favorite_genre          5000 non-null   object
dtypes: float64(3), int64(4), object(7)
memory usage: 547.0+ KB
```

```
df.head()
```

	customer_id	age	gender	subscription_type	watch_hours	last_login_days	region	device	monthly_fee	churned
0	a9b75100-82a8-427a-a208-72f24052884a	51	Other	Basic	14.73	29	Africa	TV	8.99	1
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	47	Other	Standard	0.70	19	Europe	Mobile	13.99	1
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	27	Female	Standard	16.32	10	Asia	TV	13.99	0
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	53	Other	Premium	4.51	12	Oceania	TV	17.99	1
4	4e265c34-103a-4dbb-9553-76c9aa47e946	56	Other	Standard	1.89	13	Africa	Mobile	13.99	1

```
df.isna().sum()
```

	0
customer_id	0
age	0
gender	0
subscription_type	0
watch_hours	0
last_login_days	0
region	0
device	0
monthly_fee	0
churned	0
payment_method	0
number_of_profiles	0
avg_watch_time_per_day	0
favorite_genre	0

dtype: int64

```
df = df.dropna()
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df = df.drop_duplicates()
```

```
df.shape
```

```
(5000, 14)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column                  Non-Null Count  Dtype
```

```
 ---  ------                  --------------  -----
 0   customer_id             5000 non-null   object
 1   age                     5000 non-null   int64
 2   gender                  5000 non-null   object
 3   subscription_type       5000 non-null   object
 4   watch_hours             5000 non-null   float64
 5   last_login_days         5000 non-null   int64
 6   region                  5000 non-null   object
 7   device                  5000 non-null   object
 8   monthly_fee             5000 non-null   float64
 9   churned                 5000 non-null   int64
 10  payment_method          5000 non-null   object
 11  number_of_profiles      5000 non-null   int64
 12  avg_watch_time_per_day  5000 non-null   float64
 13  favorite_genre          5000 non-null   object
dtypes: float64(3), int64(4), object(7)
memory usage: 547.0+ KB
```

Data Quality:

- 5000 customers, 14 variables
- No missing values
- No duplicate records
- Clean behavioral dataset

GDPR & Ethics:

- No sensitive personal data
- No ethnicity, religion, health, political data
- Only behavioral and subscription data used
- Model used for retention optimization
- No automated discrimination

ANALYTICAL CRM- SEGMENTATION

```
# SEGMENTATION DATA (Behavioral RFM)
# R = last_login_days
# F = watch_hours / avg_watch_time_per_day
# M = monthly_fee

seg_df = df[[
    "last_login_days",
    "watch_hours",
    "avg_watch_time_per_day",
    "monthly_fee",
    "number_of_profiles"
]].copy()

seg_df.head()
```

	last_login_days	watch_hours	avg_watch_time_per_day	monthly_fee	number_of_profiles
0	29	14.73	0.49	8.99	1
1	19	0.70	0.03	13.99	5
2	10	16.32	1.48	13.99	2
3	12	4.51	0.35	17.99	2
4	13	1.89	0.13	13.99	2

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X = scaler.fit_transform(seg_df)

X[:5]
```

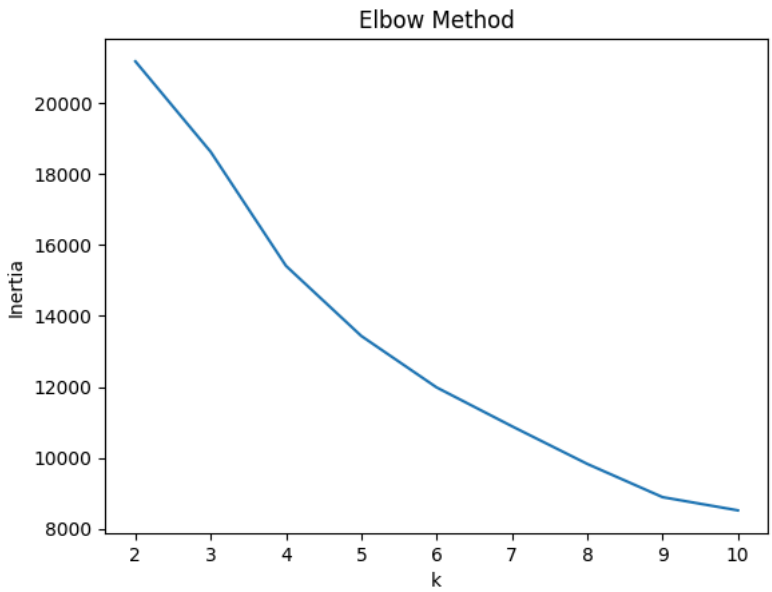
```
array([[-0.06215238,  0.25642504, -0.14689477, -1.27134085, -1.42996464],
       [-0.63246235, -0.91143244, -0.32249662,  0.08305133,  1.39549404],
       [-1.14574131,  0.38877668,  0.23103096,  0.08305133, -0.72359997],
       [-1.03167932, -0.59428796, -0.20033881,  1.16656507, -0.72359997],
       [-0.97464832, -0.81237682, -0.28432231,  0.08305133, -0.72359997]])
```

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

for k in range(2, 11):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X)
    inertia.append(km.inertia_)

plt.plot(range(2, 11), inertia)
plt.xlabel("k")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()
```




```
k = 6

km = KMeans(n_clusters=k, random_state=42)
df["segment"] = km.fit_predict(X)

df[["customer_id", "segment"]].head()
```

	customer_id	segment
0	a9b75100-82a8-427a-a208-72f24052884a	1
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	2
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	1
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	2
4	4e265c34-103a-4dbb-9553-76c9aa47e946	1

```
# Mean profile by segment
seg_profile = df.groupby("segment")[[
    "last_login_days",
    "watch_hours",
    "avg_watch_time_per_day",
    "monthly_fee",
    "number_of_profiles",
    "churned"
]].mean().round(2)

# Segment size
seg_size = df["segment"].value_counts().sort_index()

seg_profile
```

segment	last_login_days	watch_hours	avg_watch_time_per_day	monthly_fee	number_of_profiles	churned
0	30.46	8.54	0.51	8.99	3.44	0.66
1	29.24	9.24	0.56	14.20	1.30	0.62
2	15.34	7.71	0.85	16.21	3.92	0.21
3	0.85	27.60	17.36	14.08	2.94	0.00
4	30.09	40.41	2.01	13.63	3.27	0.01
5	47.31	9.52	0.20	16.18	3.73	0.69

```
seg_churn = df.groupby("segment")["churned"].mean().round(3)
seg_churn
```

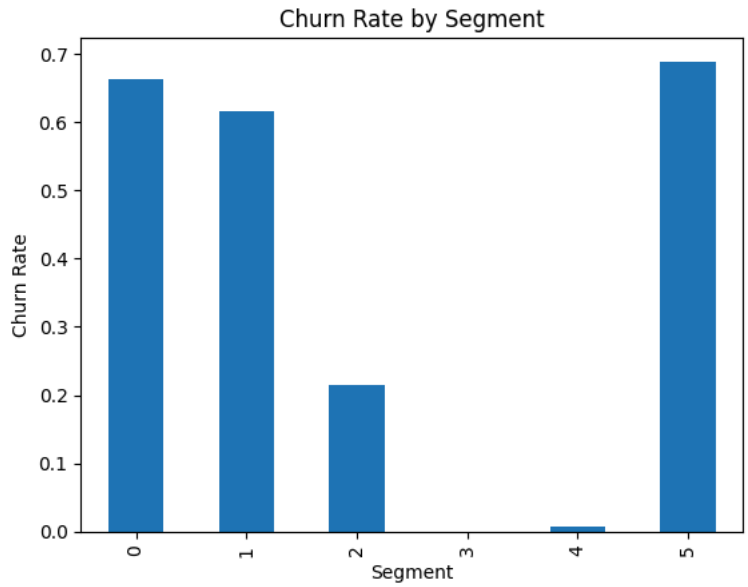
segment	churned
0	0.664
1	0.616
2	0.214
3	0.000
4	0.007
5	0.688

dtype: float64

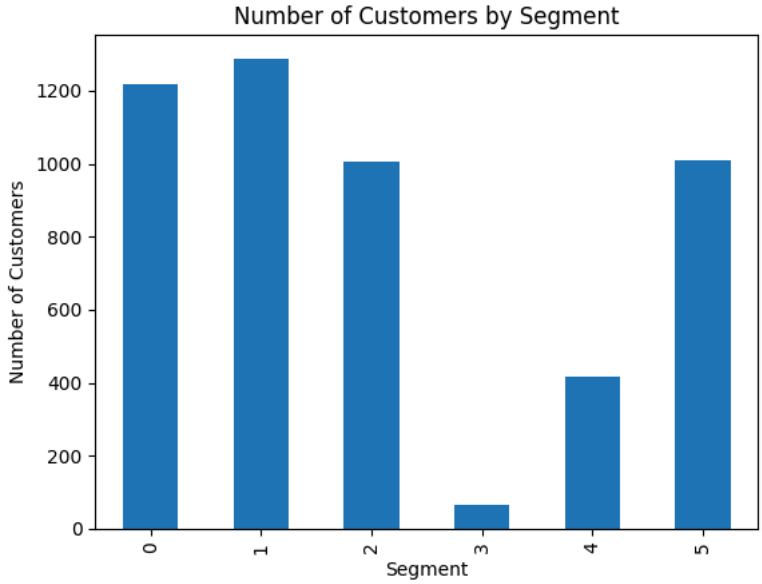
```
import matplotlib.pyplot as plt

churn_by_segment = df.groupby("segment")["churned"].mean()

plt.figure()
churn_by_segment.plot(kind="bar")
plt.title("Churn Rate by Segment")
plt.xlabel("Segment")
plt.ylabel("Churn Rate")
plt.show()
```

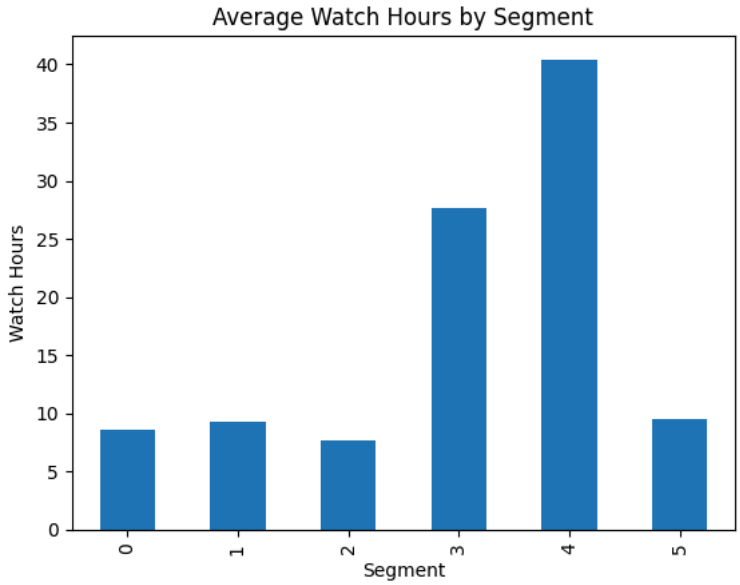



```
plt.figure()
df["segment"].value_counts().sort_index().plot(kind="bar")
plt.title("Number of Customers by Segment")
plt.xlabel("Segment")
plt.ylabel("Number of Customers")
plt.show()
```




```
watch_by_segment = df.groupby("segment")["watch_hours"].mean()

plt.figure()
watch_by_segment.plot(kind="bar")
plt.title("Average Watch Hours by Segment")
plt.xlabel("Segment")
plt.ylabel("Watch Hours")
plt.show()
```




```
summary = seg_profile.copy()
summary["customers"] = seg_size
summary["churn_rate"] = seg_churn

summary = summary[[
    "customers",
    "churn_rate",
    "last_login_days",
    "watch_hours",
    "avg_watch_time_per_day",
    "monthly_fee",
    "number_of_profiles"
]]

summary
```

segment	customers	churn_rate	last_login_days	watch_hours	avg_watch_time_per_day	monthly_fee	number_of_profi
0	1219	0.664	30.46	8.54	0.51	8.99	3
1	1288	0.616	29.24	9.24	0.56	14.20	1
2	1004	0.214	15.34	7.71	0.85	16.21	3
3	65	0.000	0.85	27.60	17.36	14.08	2
4	416	0.007	30.09	40.41	2.01	13.63	3
5	1008	0.688	47.31	9.52	0.20	16.18	3

Các bước tiếp theo: Tạo mã bằng summary New interactive sheet

```
summary["retention_rate"] = 1 - summary["churn_rate"]
summary
```

segment	customers	churn_rate	last_login_days	watch_hours	avg_watch_time_per_day	monthly_fee	number_of_profi
0	1219	0.664	30.46	8.54	0.51	8.99	3
1	1288	0.616	29.24	9.24	0.56	14.20	1
2	1004	0.214	15.34	7.71	0.85	16.21	3
3	65	0.000	0.85	27.60	17.36	14.08	2
4	416	0.007	30.09	40.41	2.01	13.63	3
5	1008	0.688	47.31	9.52	0.20	16.18	3

Các bước tiếp theo: Tạo mã bằng summary New interactive sheet

```
summary["clv_proxy"] = summary["monthly_fee"] * 12 * summary["retention_rate"]
summary
```

segment	customers	churn_rate	last_login_days	watch_hours	avg_watch_time_per_day	monthly_fee	number_of_profi
0	1219	0.664	30.46	8.54	0.51	8.99	3
1	1288	0.616	29.24	9.24	0.56	14.20	1
2	1004	0.214	15.34	7.71	0.85	16.21	3
3	65	0.000	0.85	27.60	17.36	14.08	2
4	416	0.007	30.09	40.41	2.01	13.63	3
5	1008	0.688	47.31	9.52	0.20	16.18	3

Các bước tiếp theo: Tạo mã bằng summary New interactive sheet

Segment has:

High churn + High fee → Big risk

High retention + High fees → Loyal value segment

Low fee + High churn → Low priority

```
segment_names = {
    0: "High Risk Basic",
    1: "High Risk Standard",
    2: "Stable Paying",
    3: "Top Loyal",
    4: "Heavy Streamers",
    5: "Inactive High Risk"
}

df["segment_name"] = df["segment"].map(segment_names)
df[["segment", "segment_name"]].drop_duplicates().sort_values("segment")
```

	segment	segment_name
6	0	High Risk Basic
0	1	High Risk Standard
1	2	Stable Paying
191	3	Top Loyal
15	4	Heavy Streamers
10	5	Inactive High Risk

Segment Persona Summary

Segment 0 – High Risk Basic

- High churn 66%
- Low fee, low engagement
- Price sensitive
- Low-cost retention needed

Segment 1 – High Risk Standard

- High churn 61%
- Medium fee users
- Core users at risk
- Personalized reactivation

Segment 2 – Stable Paying

- Churn 21%
- High retention
- Strong economic value
- Cross-sell opportunity

Segment 3 – Top Loyal

- Churn 0%
- Highest engagement
- Highest CLV
- VIP loyalty focus

Segment 4 – Heavy Streamers

- Very high usage
- Strong retention
- Upsell potential

Segment 5 – Inactive High Risk

- Highest churn 68%
- Long inactivity
- Immediate win-back campaign

Emotional CRM – Engagement analysis

```
# Divide engagement to watch_hours
df["engagement_level"] = pd.cut(
    df["watch_hours"],
    bins=2,
    labels=["Low", "High"]
)

df.groupby("engagement_level")["churned"].mean()
```

```
/tmp/ipython-input-951/1492708070.py:8: FutureWarning: The default of observed=False is deprecated and will be ch
  df.groupby("engagement_level")["churned"].mean()
```

engagement_level	churned
Low	0.507876
High	0.000000

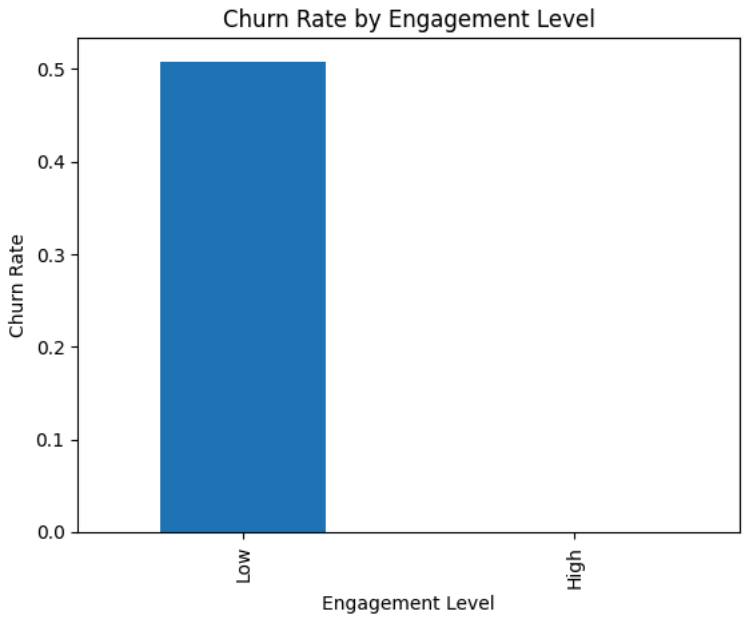
dtype: float64

```
import matplotlib.pyplot as plt

churn_by_engagement = df.groupby("engagement_level")["churned"].mean()

plt.figure()
churn_by_engagement.plot(kind="bar")
plt.title("Churn Rate by Engagement Level")
plt.xlabel("Engagement Level")
plt.ylabel("Churn Rate")
plt.show()
```

```
/tmp/ipython-input-951/1827944378.py:3: FutureWarning: The default of observed=False is deprecated and will be ch
  churn_by_engagement = df.groupby("engagement_level")["churned"].mean()
```



Low engagement users show over 50 percent churn.

High engagement users almost never churn.

Engagement is a strong churn predictor.

Strategic CRM – Churn scoring and action

```
y = df["churned"]
```

```
X_model = df[[
    "age",
    "watch_hours",
    "avg_watch_time_per_day",
    "last_login_days",
    "monthly_fee",
    "number_of_profiles"
]]
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X_model, y, test_size=0.3, random_state=42
)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
▾ LogisticRegression
LogisticRegression(max_iter=1000)
```

```
from sklearn.metrics import accuracy_score, classification_report

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.8886666666666667
              precision    recall  f1-score   support

           0       0.90      0.87      0.89       753
           1       0.87      0.91      0.89       747

    accuracy                           0.89      1500
   macro avg       0.89      0.89      0.89      1500
weighted avg       0.89      0.89      0.89      1500
```

```
df["churn_score"] = model.predict_proba(X_model)[:,1]
df[["customer_id", "churn_score"]].head()
```

	customer_id	churn_score
0	a9b75100-82a8-427a-a208-72f24052884a	0.711014
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	0.639955
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	0.003103
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	0.325962
4	4e265c34-103a-4dbb-9553-76c9aa47e946	0.811552

```
df["risk_level"] = pd.cut(
    df["churn_score"],
    bins=[0, 0.3, 0.6, 1],
    labels=["Low", "Medium", "High"]
)

df[["customer_id", "segment", "risk_level"]].head()
```

	customer_id	segment	risk_level
0	a9b75100-82a8-427a-a208-72f24052884a	1	High
1	49a5dfd9-7e69-4022-a6ad-0a1b9767fb5b	2	High
2	4d71f6ce-fca9-4ff7-8afa-197ac24de14b	1	Low
3	d3c72c38-631b-4f9e-8a0e-de103cad1a7d	2	Medium
4	4e265c34-103a-4dbb-9553-76c9aa47e946	1	High

```
overall_churn = df["churned"].mean()
overall_retention = 1 - overall_churn
avg_revenue = df["monthly_fee"].mean()

print("Overall churn rate:", round(overall_churn,3))
print("Overall retention rate:", round(overall_retention,3))
print("Average monthly revenue per user:", round(avg_revenue,2))
```

```
Overall churn rate: 0.503
Overall retention rate: 0.497
Average monthly revenue per user: 13.68
```

```
pd.crosstab(df["segment"], df["risk_level"])
```

risk_level segment	Low	Medium	High
0	336	112	771
1	381	122	785
2	668	175	161
3	65	0	0
4	415	1	0
5	202	129	677

```
kpi_by_segment = df.groupby("segment").agg(
    customers=("customer_id", "count"),
    churn_rate=("churned", "mean"),
    arpu=("monthly_fee", "mean"),
    avg_watch_hours=("watch_hours", "mean"),
    avg_watch_time_day=("avg_watch_time_per_day", "mean"),
    avg_churn_score=("churn_score", "mean")
).round(3)

kpi_by_segment
```

segment	customers	churn_rate	arpu	avg_watch_hours	avg_watch_time_day	avg_churn_score
0	1219	0.664	8.990	8.539	0.514	0.639
1	1288	0.616	14.196	9.236	0.562	0.623
2	1004	0.214	16.213	7.707	0.852	0.245
3	65	0.000	14.082	27.604	17.364	0.000
4	416	0.007	13.632	40.408	2.009	0.006
5	1008	0.688	16.180	9.525	0.203	0.684

Các bước tiếp theo: Tạo mã bằng kpi_by_segment New interactive sheet

```
import numpy as np
import pandas as pd
```

```
df["risk_level"] = pd.cut(
    df["churn_score"],
    bins=[0, 0.3, 0.6, 1.0000001],
    labels=["Low", "Medium", "High"],
    include_lowest=True
)
```

```
df["monthly_fee"] = pd.to_numeric(df["monthly_fee"], errors="coerce")

df["value_level"] = pd.cut(
    df["monthly_fee"],
    bins=[0, 12, 15, 100],
    labels=["Low", "Mid", "High"],
    include_lowest=True
)
```

```
conditions = [
    (df["risk_level"] == "High") & (df["value_level"].isin(["Mid", "High"])),
    (df["risk_level"] == "High") & (df["value_level"] == "Low"),
    (df["risk_level"] == "Low") & (df["value_level"] == "High")
]

choices = [
    "Priority 1 Retention",
    "Low-cost Retention",
    "Upsell"
]

df["action_priority"] = np.select(conditions, choices, default="Monitor")

df["action_priority"].value_counts()
```

action_priority	count
Monitor	1764
Priority 1 Retention	1414
Low-cost Retention	980
Upsell	842

dtype: int64

CRM Marketing Strategy

Priority 1 Retention

- Immediate retention campaign
- Targeted discount
- Personalized email

Low-cost Retention

- Reminder email
- Push notification

Upsell

- Premium upgrade offer
- Bundle plan

Monitor

- Regular communication
- No aggressive incentive